

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – JUNE 2012

### PY102 FUNDAMENTALS OF BIOMEDICAL PHYSICS

Date: 7.06.12, Thursday

Time: 10.00a.m to 10.20a.m

Maximum Marks: 10

#### Instructions:

1. There are two parts in the paper. PART – I contains 10 Questions and PART – II contains 4 Questions.
2. PART – I should be answered in MCQ 'S question paper & PART – II should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for PART – I is 20 min. Question paper for PART – II will be given at the table when the answer sheet for PART – I is returned. However, Candidates who complete PART – I earlier than 20 min, can collect main question paper for PART – II immediately after handing over the PART-I question paper.

#### PART-I

#### Multiple Choice Questions: (MCQs)

(10 × 1 = 10)

(Encircle the correct answer)

1. The refractive index of the core and materials of fiber optic are 1.54 and 1.5 respectively than what is the value of Numerical Aperture?
  - a. 0.526
  - b. 0.2986
  - c. 0.3487
  - d. 0.4283
2. If one mole of HCl at 298 K and 1 atm pressure occupies 24.5 liters, calculate work for the system?
  - a. 298L-atm
  - b. 24.5 L-atm
  - c. 12.16 L-atm
  - d. 0.08 L-atm
3. A construction worker uses a board and log as a lever to lift a heavy rock. If the input arm is 3 meters long and the output arm is 0.75 meters long, what is the mechanical advantage of the lever?
  - a. 2.25
  - b. 0.25
  - c. 3.75
  - d. 4
4. What is the First law of thermodynamics?
  - a.  $dQ=dU/PdV$
  - b.  $dQ=dU-PdV$
  - c.  $dQ=dU+PdV$
  - d.  $dQ = dU*PdV$

5. A nine volt battery supplies power to a cordless curling iron with a resistance of 18 ohms. How much current is flowing through the curling iron?

- a. 1A
- b. 0.5A
- c. 3A
- d. 2A

6. If frequency of emitted X-ray  $\nu = 2 \times 10^{17}$  Hz. Calculate the X-ray wavelength.

- a.  $1 \times 10^{-8}$  m
- b.  $1.5 \times 10^{-9}$  m
- c.  $2 \times 10^7$  m
- d.  $1.5 \times 10^9$  m

7. The speed and wavelength of light inside the fiber is  $2.14 \times 10^8$  m/s and  $452 \times 10^{-9}$  m. Calculate the frequency of light inside the fiber.

- a.  $4.74 \times 10^{14}$  Hz
- b.  $1.12 \times 10^{13}$  Hz
- c.  $5.55 \times 10^{10}$  Hz
- d.  $3.47 \times 10^{11}$  Hz

8. A person produces sound and this sound is reflected back from an obstacle at a distance x. The velocity of sound  $v=360$  m/s. the time taken by the sound to travel to the obstacle and back is  $t=0.1$ . Calculate the distance between the person and obstacle x.

- a. 18m
- b. 24m
- c. 30m
- d. 20m

9. The ripple factor in full wave rectifier is 0.48 and efficiency is  $\sim 81\%$ .

- a. Truth
- b. False

10. What is the unit of Numerical Aperture?

- a. Degree
- b. eV
- c. minute
- d. unit less

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|--------------------------------------------------------------------|-----------------|
| CHAROTAR UNIVERSITY OF SCI. & TECHNOLOGY<br>CHARUSAT CAMPUS CHANGA |                 |
| Examination Semester                                               | _____           |
| Subject                                                            | _____ No. _____ |
| Candidate's Seat No.                                               | _____           |
| Section                                                            | _____           |
| Date                                                               | _____           |
| Sr. Supervisor's<br>Sign.                                          |                 |



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## Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – JUNE 2012

### PY102 FUNDAMENTALS OF BIOMEDICAL PHYSICS

Date: 7.06.12, Thursday

Time: 10.20a.m to 1.00p.m

Maximum Marks: 60

#### Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 2 Hr 40min.
5. Draw the figure wherever necessary.

#### PART-II

##### SECTION –I

**Q.1. Long Question (1 out of 2)**

(1× 10= 10)

Explain LASER in details.

OR

- (i) Explain Kirchhoff's law of heat radiation.
- (ii) Explain Newton's law of cooling and describe the estimation of specific heat of liquid.

**Q.2. Write short notes on: (4 out of 5)**

(4× 5 = 20)

- a. Explain Interference and polarization.
- b. Give the definition of
  - (i) Thermal capacity
  - (ii) Water equivalent
  - (iii) Give Grouthus law
  - (iv) Give Celsius statement for 2<sup>nd</sup> law of thermodynamics.
  - (v) Emissive power
- c. Explain Piezo electric oscillator.
- d. Give velocity of sound in Air by resonance method.
- e. Write a note on friction.

##### SECTION –II

**Q.3. Long Question (1 out of 2)**

(1× 10= 10)

What is rectifier? Explain half wave rectifier, full wave rectifier and bridge rectifier.

OR

Explain shortwave diathermy, diapulse therapy and microwave diathermy in details.

**Q.4. Write short notes on: (4 out of 5)**

(4× 5 = 20)

- a. A laser source emits a light of wavelength  $0.621\mu\text{m}$  and has an output of 35mW. Calculate how many photons are emitted per minute by this laser source.
- b. Write note on cathode ray oscilloscope.
- c. Explain Thomson effect with first and second law of electrolysis.
- d. Give a detail explanation on Induction coil.
- e. Explain Zener diode as a voltage regulator.